

Gangmin Yu

Mechanical Engineering · Undergraduate Researcher

Kyonggi University · RVLab

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RESEARCH INTERESTS

Autonomous driving and navigation, vehicle dynamics and control, radar/sensor perception, robot simulation with Isaac Sim, robot mechanical design, embedded and mechatronic systems.

EDUCATION

Kyonggi University, Suwon, Republic of Korea

Mar. 2021 – Present

B.S. in Mechanical Engineering

Undergraduate Researcher, RVLab (Advisor: Prof. Minseong Choi)

GPA: 4.27 / 4.5 Major GPA: 4.41 / 4.5

English: TOEIC Speaking IH (150)

PUBLICATIONS

International Journals (On going)

Gangmin Yu* and Minseong Choi. “Vehicle Pose Estimation via License Plate-Based Camera-Radar Calibration” (*On going*) *IEEE Sensors*

Domestic Conferences

이충헌, 유강민, 박영상, 최민성, 조우성. “Swerve 구동 휴머노이드를 위한 Holonomic 실내 자율 탐색 알고리즘” *제어·로봇·시스템학회(ICROS) 학술대회*, 2026.

김도윤, 강민수, 유강민, 서형태, 박영상, 최민성. “IsaacSim 기반 휴머노이드 ACT 학습제어를 위한 VR 원격조작 및 Jitter 저감 기법” *제어·로봇·시스템학회(ICROS) 학술대회*, 2026.

성민준, 유강민, 김준규, 한승호, 양승훈, 최민성. “도로 노면 의미론적 분할 기법의 소수 클래스 불균형 해소를 위한 원근 변환 기반 데이터 증강 기법” *제어·로봇·시스템학회(ICROS) 학술대회*, 2026.

RESEARCH & PROJECTS

ONGOING

4D Radar Ego-Velocity Estimation

RVLab, Kyonggi University

- Estimating ego-vehicle velocity directly from 4D radar measurements toward robust self-motion estimation for autonomous driving.

Manipulation & Navigation for Humanoids in Domestic Environments

RVLab

- Applied physics-based digital-twin simulation and MetaQuest 3 teleoperation in Isaac Sim.
- Implemented autonomous driving and navigation using NAV2 on a generated 2D LiDAR map.

PAST

3D Simulation Environments & Physics Modeling for Daily-Life Scenes

KIMM (commissioned)

funded by Korea Institute of Machinery & Materials (KIMM)

- Built physically simulated indoor environments in NVIDIA Isaac Sim (PhysX) for realistic robot interaction.
- Authored rigid-body and collider models from .obj meshes; set per-object mass, inertia, friction, restitution.

- Defined articulated structures (hinge/prismatic joints) for doors and drawers with motion limits and dynamics.

Sensor Mount Design for ROBOTIS AI Worker

RVLab · Humanoid Hardware

- Designed a torso-mounted bracket to integrate an additional 3D LiDAR.
- Designed an angle-adjustable gripper camera mount to improve field of view during grasping.

Autonomous Vehicle Build

Autonomous Driving Club

- As club leader, oversaw the entire vehicle build process, designing the hardware/embedded system (actuators, sensors, MCU communication).
- Implemented steering control (Pure Pursuit) and speed control (PID).

High-Voltage Shutdown Circuit

Self-Built Car Club · Formula (EV)

- Designed a safety shutdown circuit for 120 V high voltage, cutting power on BMS/IMD/interlock faults or sudden-acceleration detection.
- Designed and fabricated the PCB (KiCad) and applied it to the vehicle.

Braking System Design

Self-Built Car Club · Formula & Baja

- Computed braking force from vehicle kinematics and designed the brake disc accordingly.
- Used CAE to optimize the design and verify structural safety.

AWARDS

Excellence Award — Self-Built Autonomous Vehicle Division (Leader)

Nov. 2025

University Creative Mobility Competition · Korea Transportation Safety Authority (KOTSA)

Acceleration Award — Baja Student (Team Member)

Nov. 2021

International Self-Built Car Competition · Yeungnam University

TECHNICAL SKILLS

Simulation / Robotics: NVIDIA Isaac Sim, PhysX, ROS2, AMCL / Navigation

Control / Embedded: Embedded C, PID, Pure Pursuit, MCU & Sensors

Design / CAE: KiCad (Schematic / PCB), AutoCAD, CATIA, Ansys

Domain: Autonomous Driving, Vehicle Dynamics, Radar Perception, Mechatronics

Last updated: June 2026